

COUNTEREXAMPLES TO WIRDANE'S CYCLOTOMIC CATALAN CONJECTURE

ABSTRACT. We disprove the literal form of Wirdane's Cyclotomic Catalan Conjecture, Conjecture 6.6 in arXiv:2605.14682v1. For nontrivial cyclotomic orders, the first counterexample in the parameter n is $(n, \mu) = (2, 3)$. We also give three infinite families of counterexamples: $(2m+1, 2)$ for $m \geq 1$, $(3r+2, 3)$ for $r \geq 0$, and $(p-1, p)$ for every odd prime p .

Let $C_0(q) = 1$ and, for $n \geq 1$, let

$$(1) \quad C_n(q) = \sum_{k=0}^{n-1} q^k C_k(q) C_{n-1-k}(q)$$

be the F rlinger–Hofbauer q -Catalan numbers [1]. Write

$$\text{Cat}_m = \frac{1}{m+1} \binom{2m}{m}, \quad \mathcal{C}(t) = \sum_{m \geq 0} \text{Cat}_m t^m = 1 + t\mathcal{C}(t)^2.$$

At $q = 1$, recurrence (1) is the Catalan recurrence, so $C_n(1) = \text{Cat}_n$.

Fix $\mu \geq 2$ and $\zeta_\mu = e^{2\pi i/\mu}$. We first recall the objects in terms of which Conjecture 6.6 of [2] is stated. Let $S_n(312)$ be the set of 312-avoiding permutations of $\{1, \dots, n\}$, let $S_{n,k}(312) = \{\pi \in S_n(312) : \pi_{k+1} = n\}$ for $0 \leq k \leq n$, and let $S'_{n,k}(312) = \bigsqcup_{j=0}^k S_{n,j}(312)$. The value n occupies exactly one position of each $\pi \in S_n(312)$, so $S_{n,0}(312), \dots, S_{n,n-1}(312)$ partition $S_n(312)$ while $S_{n,n}(312) = \emptyset$; hence

$$(2) \quad S'_{n,n}(312) = S_n(312).$$

For $1 \leq i \leq \mu$, the i -inversions of π are counted by

$$\text{inv}_i(\pi) = \#\{(a, b) : 1 \leq a < b \leq n, \pi_a > \pi_b, \pi_a \equiv i \pmod{\mu}\}$$

[2, Definition 6.1], and the multivariate q -Catalan triangle is

$$C_{n,k}(q_1, \dots, q_\mu) = \sum_{\pi \in S'_{n,k}(312)} \prod_{i=1}^{\mu} q_i^{\text{inv}_i(\pi)}$$

[2, Definition 6.2], its cyclotomic analogue being $C_{n,k}^{(\mu)} = C_{n,k}(\zeta_\mu, \dots, \zeta_\mu)$. Because $1, \dots, \mu$ is a complete residue system modulo μ , each inversion

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(a, b) of π is counted by exactly one of the inv_i , so $\sum_{i=1}^{\mu} \text{inv}_i(\pi) = \text{inv}(\pi)$, the inversion number. With (2) this gives

$$(3) \quad C_{n,n}^{(\mu)} = \sum_{\pi \in S_n(312)} \zeta_{\mu}^{\sum_{i=1}^{\mu} \text{inv}_i(\pi)} = \sum_{\pi \in S_n(312)} \zeta_{\mu}^{\text{inv}(\pi)} = C_n(\zeta_{\mu}),$$

the last equality being the interpretation

$$C_n(q) = \sum_{\pi \in S_n(312)} q^{\text{inv}(\pi)}$$

of [1], recorded in [2] as equation (2) and again as Remark 3.4. Equivalently, one may specialize all q_i to a common q by [2, Corollary 6.5] and then invoke [2, Theorem 3.3], namely $C_{n,k}(q) = \sum_{\pi \in S'_{n,k}(312)} q^{\text{inv}(\pi)}$. We do not use [2, Proposition 2.4], which asserts $C_{n,n}(q) = C_n(q)$ outright, its proof being deferred there to a work in preparation. Conjecture 6.6 of [2] reads, in full: “If $\mu \mid (n+1)$: $C_{n,n}^{(\mu)} = \frac{1}{\mu} \binom{n+1}{(n+1)/\mu}$.” By (3) it asserts that, whenever $\mu \mid n+1$,

$$(4) \quad C_n(\zeta_{\mu}) = \frac{1}{\mu} \binom{n+1}{(n+1)/\mu}.$$

As printed the conjecture places no restriction on μ , and the case $\mu = 1$ is discussed in [2, Remark 6.4]; we standardize on $\mu \geq 2$ because $\mu = 1$ is a degenerate reading of “cyclotomic”, and dispose of $\mu = 1$ separately in the closing Remark.

Theorem 1. *Equation (4) is false. Each of the following is an infinite family of counterexamples:*

$$(n, \mu) = (2m+1, 2) \quad (m \geq 1), \quad (n, \mu) = (3r+2, 3) \quad (r \geq 0),$$

$$(n, \mu) = (p-1, p) \quad (p \text{ an odd prime}).$$

Among $\mu \geq 2$, the least value of n occurring in a counterexample is $n = 2$, with $\mu = 3$.

Proof. All generating functions below are formal power series. Set

$$F_q(z) = \sum_{n \geq 0} C_n(q) z^n.$$

Recurrence (1) is equivalent to

$$(5) \quad F_q(z) = 1 + zF_q(z)F_q(qz).$$

Order two. Put $F(z) = F_{-1}(z)$. Applying (5) at z and $-z$ gives

$$F(z) = 1 + zF(z)F(-z), \quad F(-z) = 1 - zF(-z)F(z),$$

so $F(z) + F(-z) = 2$. Thus, for $H(z) = F(z) - 1$,

$$H(z) = z(1 - H(z)^2).$$

The unique solution with zero constant term is $H(z) = z\mathcal{C}(-z^2)$. Consequently,

$$(6) \quad C_{2m}(-1) = 0 \quad (m \geq 1), \quad C_{2m+1}(-1) = (-1)^m \text{Cat}_m \quad (m \geq 0).$$

For $n = 2m + 1$, the conjectured value is

$$\frac{1}{2} \binom{2m+2}{m+1} = (2m+1) \text{Cat}_m,$$

which differs from (6) for every $m \geq 1$.

Order three. Let $\omega = \zeta_3$ and put $A(z) = \mathcal{C}(z^3)$. Define

$$G(z) = \frac{1 + \omega z^2 A(z)}{1 - z}.$$

Since $A(z) = 1 + z^3 A(z)^2$ and $1 + \omega + \omega^2 = 0$,

$$(1 + \omega z A(z))(1 - \omega z) = (1 + \omega z^2 A(z))(1 + z^2 A(z)).$$

Moreover,

$$G(\omega z) = \frac{1 + z^2 A(z)}{1 - \omega z},$$

so the displayed identity gives $G(z) - 1 = zG(z)G(\omega z)$. The functional equation (5) determines its coefficients recursively, so

$$(7) \quad F_\omega(z) = \frac{1 + \omega z^2 \mathcal{C}(z^3)}{1 - z}.$$

Therefore, for $r \geq 0$,

$$(8) \quad C_{3r+2}(\omega) = 1 + \omega \sum_{j=0}^r \text{Cat}_j.$$

Its imaginary part is positive, whereas the right side of (4) is real. Thus every $(3r+2, 3)$ is a counterexample. In particular,

$$C_2(\omega) = 1 + \omega \neq 1 = \frac{1}{3} \binom{3}{1}.$$

Odd prime orders. Let p be an odd prime, $\zeta = \zeta_p$, and $\mathfrak{p} = (1 - \zeta) \subset \mathbb{Z}[\zeta]$. Since $\Phi_p(1) = p$,

$$\mathbb{Z}[\zeta]/\mathfrak{p} \cong \mathbb{F}_p, \quad \zeta \equiv 1 \pmod{\mathfrak{p}}.$$

Hence

$$C_{p-1}(\zeta) \equiv C_{p-1}(1) = \text{Cat}_{p-1} \pmod{\mathfrak{p}}.$$

Moreover,

$$(2p-1) \text{Cat}_{p-1} = \binom{2p-1}{p-1} = \prod_{j=1}^{p-1} \frac{p+j}{j} \equiv 1 \pmod{p},$$

so $\text{Cat}_{p-1} \equiv -1 \pmod{p}$. Therefore

$$C_{p-1}(\zeta_p) \equiv -1 \not\equiv 1 \pmod{\mathfrak{p}},$$

the two residues being distinct because p is odd. The value predicted by (4) at $(n, \mu) = (p-1, p)$ is $\frac{1}{p} \binom{p}{1} = 1$, proving the prime family.

Finally, no $\mu \geq 2$ divides $n+1$ when $n=0$. When $n=1$, the only admissible order is $\mu=2$, and both sides of (4) equal 1. Thus $(2, 3)$ has the least possible value of n . $\square$

Remark. The three families are far from exhausting the failures. Among the 81 pairs (n, μ) with $\mu \geq 2$, $\mu \mid n+1$ and $n \leq 29$, reducing $C_n(q)$ modulo the cyclotomic polynomial $\Phi_\mu(q)$ shows that (4) holds at exactly one of them, $(n, \mu) = (1, 2)$, and fails at the remaining 80; the three families of Theorem 1 account for 32 of the 81. The order-three argument is also the cheapest of the three, using only that the left side of (4) is not real while the right side is; that observation alone disposes of 66 of the 81 pairs, against the 9 reached by the congruence for odd prime orders.

Remark. For $\mu=2$, Conjecture 6.6 is in conflict with an assertion made one line later in [2]. Remark 6.7 there states: “For $\mu=2$: $C_{n,k}^{(2)} = (-1)^{\lfloor n/2 \rfloor} C_{n,k}$ (verified for $n \leq 5$).” Here $C_{n,k}$ is the classical Catalan triangle, obtained by setting every $q_i = 1$ in Definition 6.2 [2, Corollary 6.5]; the computation giving (3) then yields $C_{n,n} = C_n(1) = \text{Cat}_n$ on the diagonal, so Remark 6.7 asserts that $C_n(-1) = (-1)^{\lfloor n/2 \rfloor} \text{Cat}_n$. This is false for every even $n \geq 2$, because $\text{Cat}_n \neq 0$ whereas $C_n(-1) = 0$ by (6); the smallest instance is $n=2$, where it predicts $-\text{Cat}_2 = -2$ against $C_2(-1) = 0$. At $n=k=3$ three values collide: the true one is $C_{3,3}^{(2)} = C_3(-1) = -1$, visible both from (6) and from the value $C_{3,3}(q) = 1+2q+q^2+q^3$ tabulated in [2]; Remark 6.7 predicts $-\text{Cat}_3 = -5$; and Conjecture 6.6 predicts $\frac{1}{2} \binom{4}{2} = 3$. The $\mu=2$ branch of Conjecture 6.6 is therefore already contradicted inside [2], independently of the counterexamples above.

Remark. Reference [2] commits to the Carlitz–Fürlinger–Hofbauer normalization, through its equation (1), which is (1), and through its Theorem 3.3 and Proposition 2.4. Take instead the Gaussian q -Catalan number

$$\widetilde{\text{Cat}}_n(q) = \frac{1}{[n+1]_q} \binom{2n}{n}_q, \quad [j]_q = 1 + q + \cdots + q^{j-1},$$

where $\binom{2n}{n}_q$ is the Gaussian binomial coefficient. Then the right-hand side of (4) is reproduced exactly at $\mu=2$: one has $\widetilde{\text{Cat}}_n(-1) = \frac{1}{2} \binom{n+1}{(n+1)/2}$ for every odd $n \leq 11$, the common values being 1, 3, 10, 35, 126, 462. At order three, by contrast, $\widetilde{\text{Cat}}_2(q) = 1+q^2$, so $\widetilde{\text{Cat}}_2(\zeta_3) = 1+\zeta_3^2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i \neq 1$.

Remark. The theorem concerns the literal formulation of Conjecture 6.6 in arXiv:2605.14682v1, with the cyclotomic specialization derived directly from Definitions 6.1 and 6.2 as in (3); no other specialization assertion in [2] is used. The theorem makes no assertion about a later revised formulation. If the trivial order $\mu=1$ is admitted, the same formula already fails at $n=2$.

The case $p = 3$ in the prime family coincides with the first member of the order-three family.

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